

# Transactions

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## ACID Properties

- **Atomicity** - All statements succeed or none succeed.
- **Consistency** - Data moves from one valid state to another.
- **Isolation** - Parallel transactions don't interfere.
- **Durability** - Committed data is permanently saved.

# Transactions

*Disable autocommit*

**SET** autocommit = 0;

*Enable autocommit*

**SET** autocommit = 1;

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# Transactions

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## Start & Commit

**START TRANSACTION;**

**UPDATE** *accounts* **SET** *balance = balance - 50* **WHERE** *id = 1;*

**UPDATE** *accounts* **SET** *balance = balance + 50* **WHERE** *id = 2;*

**COMMIT;**

# Transactions

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## Rollback

**START TRANSACTION;**

**UPDATE** *accounts* **SET** *balance = balance - 100* **WHERE** *id = 1;*

**UPDATE** *accounts* **SET** *balance = balance + 100* **WHERE** *id = 3;*

**ROLLBACK;**

# Transactions

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## Savepoint

**START TRANSACTION;**

**UPDATE** *accounts* **SET** *balance = balance + 1000* **WHERE** *id = 1;*

**SAVEPOINT** *after\_wallet\_topup;*

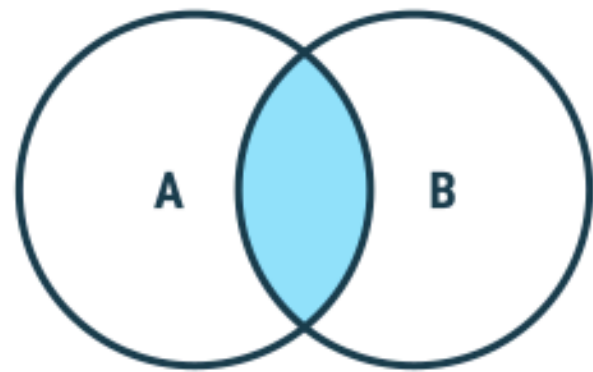
**UPDATE** *accounts* **SET** *balance = balance + 10* **WHERE** *id = 1;*

**ROLLBACK TO** *after\_wallet\_topup;*

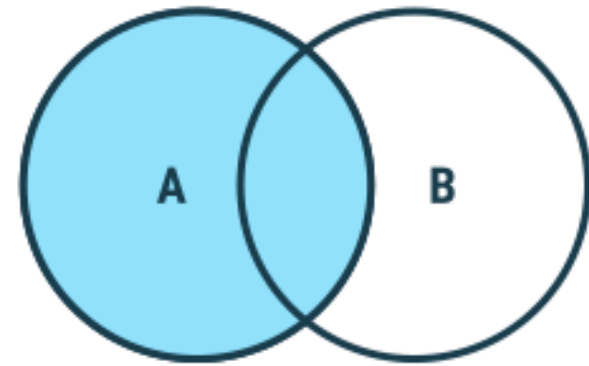
**COMMIT;**

# JOINS

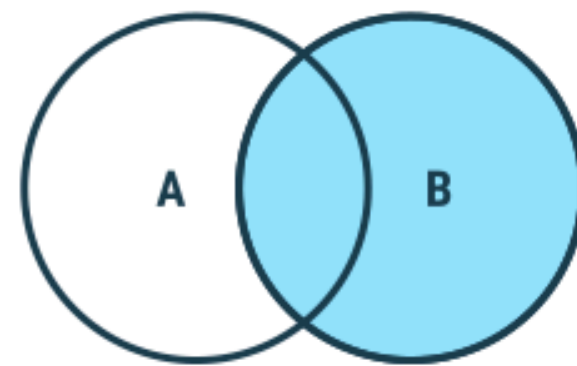
JOINS are used to combine rows from two or more tables based on a related column between them.



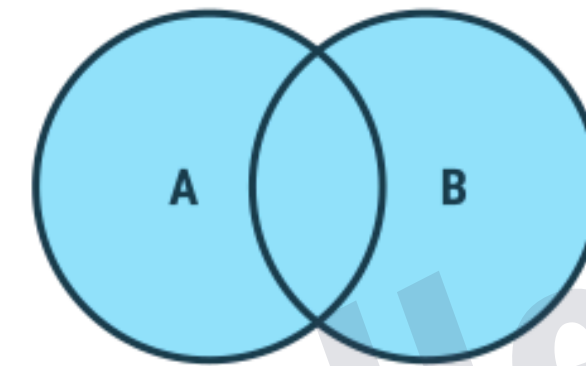
Inner Join



Left Join



Right Join

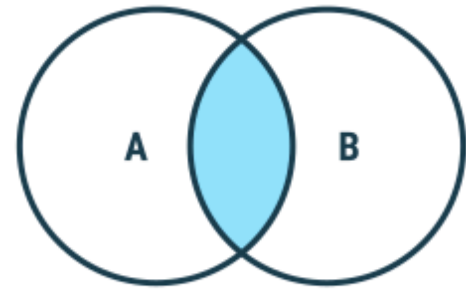


Full Join

Outer Joins

# JOINS

## INNER JOIN



| customer_id | name    | city      |
|-------------|---------|-----------|
| 1           | Alice   | Mumbai    |
| 2           | Bob     | Delhi     |
| 3           | Charlie | Bangalore |
| 4           | David   | Mumbai    |

*customers*

| order_id | customer_id | amount |
|----------|-------------|--------|
| 101      | 1           | 500    |
| 102      | 1           | 900    |
| 103      | 2           | 300    |
| 104      | 5           | 700    |

*orders*

### Syntax

**SELECT** *column(s)*

**FROM** *tableA*

**INNER JOIN** *tableB*

**ON** *tableA.col\_name = tableB.col\_name;*

```
-- inner join
```

```
SELECT c.name, o.order_id, o.amount
```

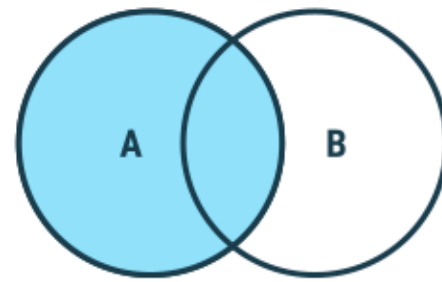
```
FROM customers c
```

```
INNER JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

# JOINS

## LEFT JOIN



| customer_id | name    | city      |
|-------------|---------|-----------|
| 1           | Alice   | Mumbai    |
| 2           | Bob     | Delhi     |
| 3           | Charlie | Bangalore |
| 4           | David   | Mumbai    |

*customers*

| order_id | customer_id | amount |
|----------|-------------|--------|
| 101      | 1           | 500    |
| 102      | 1           | 900    |
| 103      | 2           | 300    |
| 104      | 5           | 700    |

*orders*

### Syntax

**SELECT** *column(s)*

**FROM** *tableA*

**LEFT JOIN** *tableB*

**ON** *tableA.col\_name = tableB.col\_name;*

```
-- left join
```

```
SELECT *
```

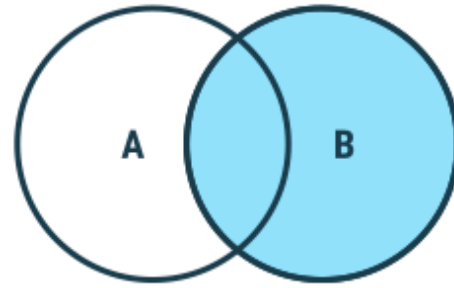
```
FROM customers c
```

```
LEFT JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

# JOINS

## RIGHT JOIN



| customer_id | name    | city      |
|-------------|---------|-----------|
| 1           | Alice   | Mumbai    |
| 2           | Bob     | Delhi     |
| 3           | Charlie | Bangalore |
| 4           | David   | Mumbai    |

*customers*

| order_id | customer_id | amount |
|----------|-------------|--------|
| 101      | 1           | 500    |
| 102      | 1           | 900    |
| 103      | 2           | 300    |
| 104      | 5           | 700    |

*orders*

### Syntax

**SELECT** *column(s)*

**FROM** *tableA*

**RIGHT JOIN** *tableB*

**ON** *tableA.col\_name = tableB.col\_name;*

```
-- right join
```

```
SELECT *
```

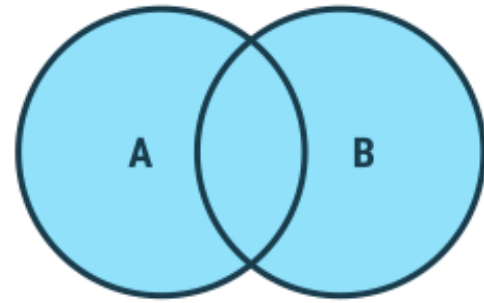
```
FROM customers c
```

```
RIGHT JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

# JOINS

## OUTER JOIN



*LEFT JOIN*

*UNION*

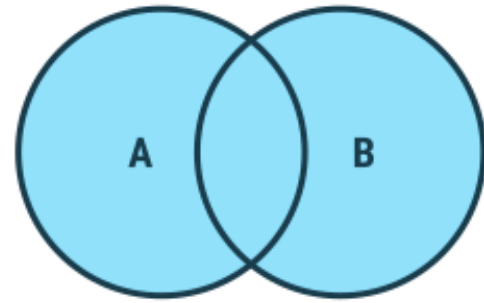
*RIGHT JOIN*

## *Syntax in MySQL*

```
SELECT * FROM customers as c
LEFT JOIN orders as o
ON c.customer_id = o.customer_id
UNION
SELECT * FROM customers as c
RIGHT JOIN orders as o
ON c.customer_id = o.customer_id;
```

# JOINS

## CROSS JOIN



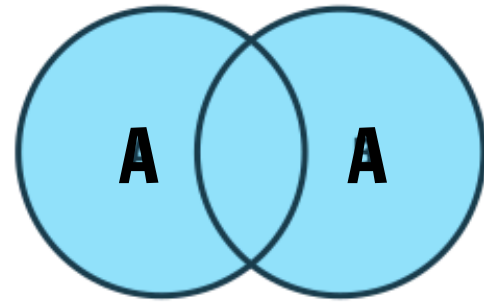
### Syntax

```
SELECT column(s)  
FROM tableA  
CROSS JOIN tableB ;
```

```
-- cross join  
SELECT *  
FROM customers as c  
CROSS JOIN orders as o;  
  
-- inner join  
SELECT *  
FROM customers as A  
JOIN customers as B  
ON A.customer_id = B.customer_id;
```

# JOINS

## SELF JOIN



It is a regular join but the table is joined with itself.

### *Syntax*

**SELECT** *column(s)*

**FROM** *table as a*

**JOIN** *table as b*

**ON** *a.col\_name = b.col\_name;*

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# Practice Qs

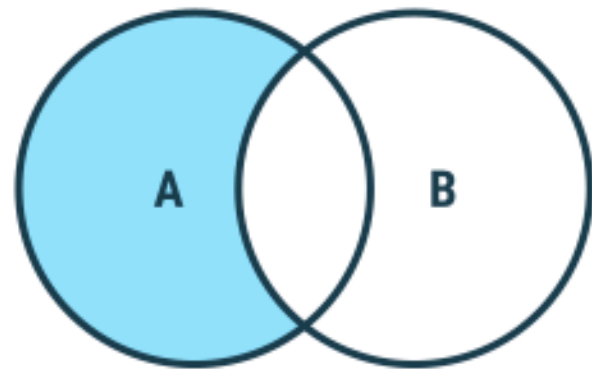
Write SQL command to display the exclusive joins :

| customer_id | name    | city      |
|-------------|---------|-----------|
| 1           | Alice   | Mumbai    |
| 2           | Bob     | Delhi     |
| 3           | Charlie | Bangalore |
| 4           | David   | Mumbai    |

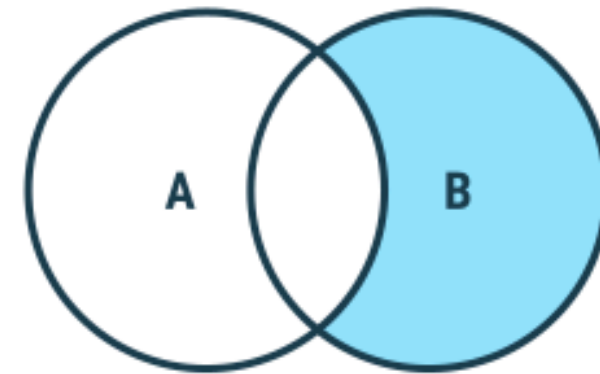
customers

| order_id | customer_id | amount |
|----------|-------------|--------|
| 101      | 1           | 500    |
| 102      | 1           | 900    |
| 103      | 2           | 300    |
| 104      | 5           | 700    |

orders



Left Exclusive Join



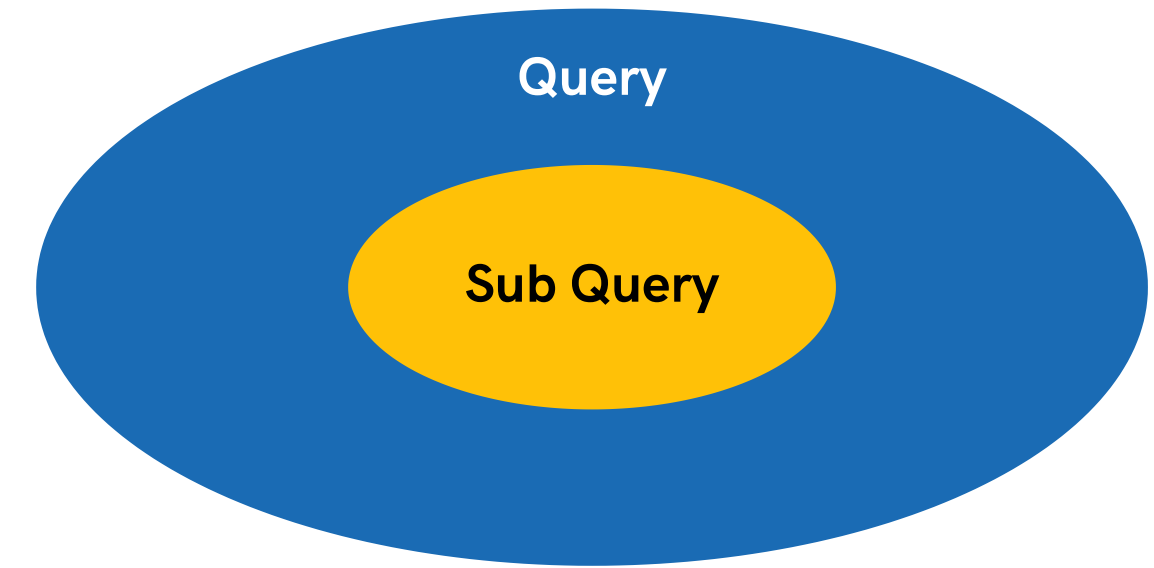
Right Exclusive Join

```
-- left exclusive
SELECT *
FROM customers as c
LEFT JOIN orders as o
ON c.customer_id = o.customer_id
WHERE o.customer_id IS NULL;
```

```
-- right exclusive
SELECT *
FROM customers as c
RIGHT JOIN orders as o
ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL;
```

# Sub-Queries

A Subquery or Inner query or a Nested query is a query within another SQL query. It involves 2 select statements.



## *Syntax*

**SELECT** *column(s)*

**FROM** *table\_name*

**WHERE** *col\_name operator*

( *subquery* );

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# Sub-Queries

---

With WHERE

```
SELECT *  
FROM orders  
WHERE amount > (  
    SELECT AVG(amount)  
    FROM orders  
);
```

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# Sub-Queries

---

With SELECT

```
SELECT name,  
    ( SELECT COUNT(*)  
      FROM orders o  
      WHERE o.customer_id = c.customer_id  
    ) as order_count  
FROM customers c;
```

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# Sub-Queries

---

With FROM

```
SELECT
    summary.customer_id,
    summary.avg_amount
FROM
    (
        SELECT
            customer_id,
            AVG(amount) AS avg_amount
        FROM orders
        GROUP BY customer_id
    ) AS summary;
```

# Views in SQL

**A view is a virtual table based on the result-set of an SQL statement.**

## *Syntax*

```
CREATE VIEW view1 AS  
SELECT col1, col2 FROM table_name;
```

\*A view always shows up-to-date data.  
The database engine recreates the view,  
every time a user queries it.

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# Views in SQL

- No data is stored physically (unless it's a materialized view in some DBs).
- Can include columns from one or more tables.
- Can be used in SELECT, JOIN, or even WHERE clauses like a normal table.
- Helps with security by exposing only certain columns to users.

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# Index in SQL

indexes are special database objects that make **data retrieval faster**.

*Syntax (single col & multi-col)*

**CREATE** INDEX *idx\_name* ON *table*(*col*);

**CREATE** INDEX *idx\_name* ON *table*(*col1*, *col2*);

**SHOW** INDEX FROM *table*;

**DROP** INDEX *idx\_name* ON *table*;

# Stored Procedures

Predefined set of SQL statements that you can save in the database and execute whenever needed.

*Syntax (Create)*

**CREATE PROCEDURE** *procedure\_name* (*parameters*)

**BEGIN**

-- SQL statements

**END;**

```
DELIMITER $$
```

```
CREATE PROCEDURE check_balance(IN acc_id INT, OUT bal DECIMAL(10, 2))
```

```
BEGIN
```

```
    SELECT balance INTO bal
```

```
    FROM bank_accounts as b
```

```
    WHERE b.account_id = acc_id;
```

```
END $$
```

```
DELIMITER ;
```

```
CALL check_balance(2, @balance);
```

```
SELECT @balance;
```

# Stored Procedures

## *Syntax (Call)*

**CALL** *procedure\_name* (*arguments*);

## *Syntax (Drop)*

**DROP** PROCEDURE **IF EXISTS** *procedure\_name*;